University of Khartoum

Faculty of Engineering

Department of Electrical and Electronics Engineering

AI Multi-Agent Anti-UAV Detection System

A Distributed Edge-Computing Framework

for Real-Time Anti-UAV Detection and Tracking

Project Report — Updated April 2026

Authors:

Abdelsamad Abdalla Abdelsamad
Mustafa Mubarak Elsaeed

Supervisor:

D. Magdi Amen

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Contents

Abstract

This report presents the complete design, implementation, and experimental evaluation of an AI-powered distributed anti-UAV detection system. The system combines a YOLO26s object detection model running on edge devices with a centralised control center accessible via web browser and mobile devices.

The production model, **BirdDrone-2C-FT**, is a fine-tuned 2-class Bird/Drone YOLO26s detector trained on curated aerial imagery and subsequently fine-tuned on pseudo-labeled frames from the DUT Anti-UAV benchmark. It achieves mAP@0.5 = 0.969 on the combined validation set, an average detection rate of 0.818 across 20 DUT benchmark videos, and a bird false alarm rate of just 0.3%. Fine-tuning reduces low-confidence false positives by 55% compared to the base model, with negligible regression on original validation sets.

The control center (v2) provides JWT-based authentication with invite-token registration, a real-time dashboard with per-class detection color coding, an interactive map with slide-in device panels, WebRTC live video streaming, and edge device health monitoring.

Chapter 1

Introduction

The proliferation of consumer and commercial unmanned aerial vehicles (UAVs) poses increasing challenges to airspace security, critical infrastructure protection, and aviation safety. Vision-based anti-UAV systems must reliably detect aerial objects in real time, distinguishing between genuine threats (drones) and confuser objects (birds) that share similar visual characteristics at distance.

This project addresses the full pipeline from dataset curation and model training to real-time deployment on distributed edge hardware. The system architecture follows a two-tier design: edge devices running YOLO26s inference on local camera feeds, publishing detection results over MQTT to a central control center that aggregates, visualises, and allows operators to control all connected devices.

The chosen production model is **BirdDrone-2C-FT** — a YOLO26s model trained on a curated 2-class (Bird/Drone) dataset and fine-tuned on the DUT Anti-UAV benchmark. This report documents the full training methodology, evaluation results, and deployment architecture for this model.

1.1 Objectives

1. Train a YOLO26s model for 2-class aerial object detection (Bird, Drone) on curated RGB datasets with balanced class representation.
2. Fine-tune the model on the DUT Anti-UAV benchmark to improve real-world detection continuity and reduce false positives.
3. Implement a distributed edge-computing inference pipeline with MQTT communication, WebRTC live streaming, and PTZ camera control.

4. Develop a modern web-based control center with authentication and real-time monitoring.

1.2 Scope and Limitations

- Training datasets are limited to publicly available RGB drone and bird imagery (CC BY 4.0 and CC0 licenses).
- Thermal infrared modality is identified as future work.
- The DUT Anti-UAV evaluation uses proxy metrics (detection rate, tracking gaps) since per-frame annotations are not available in the tracking split.
- The WOSDETC challenge dataset requires a data usage agreement and was not used in this work.

1.3 Report Structure

Chapter 2 reviews related work. Chapter 3 describes the full methodology. Chapter 4 presents training and evaluation results. Chapter 5 discusses the findings. Chapter 6 concludes with recommendations.

Chapter 2

Literature Review

2.1 YOLO Model Family

The YOLO (You Only Look Once) family of single-stage object detectors has dominated real-time detection benchmarks since its introduction. YOLO26 [1], released by Ultralytics in September 2025, introduces several key innovations relevant to aerial UAV detection:

- **NMS-free end-to-end inference**: eliminates non-maximum suppression post-processing, reducing deployment latency.
- **MuSGD optimizer**: momentum-updated SGD with improved convergence on small and imbalanced datasets.
- **STAL (Small-Target-Aware Label Assignment)**: improves detection of small objects, directly addressing distant drones.
- **DFL removal**: simplified distribution focal loss reduces model complexity without sacrificing accuracy.

YOLO26s (9.9M parameters, 22.5 GFLOPs) was selected for this work as it fits within the RTX 2070 8GB VRAM budget at batch 16 (4.7GB peak) while providing state-of-the-art accuracy for aerial imagery. The “s” (small) variant is also deployable on edge hardware such as Raspberry Pi 4 and NVIDIA Jetson Nano, making it suitable for the distributed inference architecture of this system.

2.2 Drone vs. Bird Detection

The Drone-vs-Bird Detection Grand Challenge (WOSDETC) [2], held annually at AVSS, ICIAP, and ICASSP, provides the most directly comparable benchmark. The 2023 edition reported top methods achieving F1 $\approx$ 0.91 (OBSS, 1st place) and 0.88 (multi-scale YOLOv8, IJCNN 2025) [5]. The challenge uses RGB video sequences with small drone targets against complex backgrounds including birds.

YOLOBirDrone [4] (arXiv:2601.08319) proposes a dedicated dataset and enhanced YOLO architecture for bird vs. drone classification, reporting improved discrimination over standard YOLO baselines. Their work confirms that the Bird/Drone boundary is the primary challenge in aerial detection systems — the same challenge that BirdDrone-2C-FT is specifically designed to address.

2.3 DUT Anti-UAV Benchmark

The DUT Anti-UAV dataset [3] (IEEE-TITS 2022) provides 20 RGB daytime UAV detection sequences with published baselines: YOLOX 0.720, Cascade-RCNN 0.680, ATSS 0.665, Faster R-CNN 0.621. The dataset is particularly challenging due to complex backgrounds (buildings, trees, sky) and small UAV apparent sizes at distance. BirdDrone-2C-FT achieves an average detection rate of 0.818 on these sequences, exceeding the published YOLOX baseline.

2.4 Transfer Learning and Fine-Tuning for Aerial Detection

Transfer learning from large-scale detection models to domain-specific aerial datasets is well-established. The key challenge is catastrophic forgetting: aggressive fine-tuning can erase generalisation learned during base training. This work addresses this by using a $10\times$ reduced learning rate (lr0 = 0.001 vs 0.01 for base training), short patience (8 epochs), and copy-paste augmentation to maintain diversity in the fine-tuning data.

2.5 Distributed Edge Inference

Stacker et al. [6] (ICCVW 2021) demonstrate that runtime optimisation techniques (quantisation, TensorRT export, batch size tuning) are critical for achieving acceptable inference throughput on resource-constrained edge hardware. This informs our model selection (YOLO26s over larger variants) and configurable frame rate design.

2.6 Distance and Trajectory Estimation

When enabled, the system estimates UAV distance using the pinhole camera model:

$$d = \frac{W_{\text{real}} \cdot f}{W_{\text{bbox}}} \quad \text{where} \quad f = \frac{W_{\text{frame}}}{2\tan(\theta_{\text{FOV}}/2)} \tag{2.1}$$

where W_{real} is the known UAV wingspan (configurable, default 0.5m), W_{bbox} is the bounding box width in pixels, and θ_{FOV} is the camera horizontal field of view. Trajectory estimation computes the mean frame-to-frame centroid displacement over a rolling window of N frames.

Chapter 3

Methodology

3.1 Canonical Class Taxonomy

BirdDrone-2C-FT uses a 2-class taxonomy:

Table 3.1: Canonical class taxonomy for BirdDrone-2C-FT.

Class	Index	Description
Bird	0	Birds — confuser class, not a threat
Drone	1	All aerial vehicles: consumer drones, quadcopters, fixed-wing UAVs

The 2-class taxonomy was chosen because bird/drone discrimination is the primary operational requirement. Collapsing all aerial vehicle types into a single "Drone" class maximises training data for the threat category and avoids annotation ambiguity between drone subtypes.

3.2 Source Datasets

All training datasets are licensed under CC BY 4.0 or CC0.

Table 3.2: Source datasets used for BirdDrone-2C base training.

Dataset	Source	License	Images	Class
Birds.v1i.yolov8 [7]	Roboflow Universe	CC BY 4.0	3,404	Bird
fixed-wing-uav [8]	Kaggle (nyahmet)	CC0	554	Drone
anti-uav [9]	Roboflow Universe	CC BY 4.0	2,850*	Drone

*Capped from 19,849 to maintain class balance

3.3 BirdDrone-2C Base Dataset (6,808 images)

The base dataset was constructed to achieve perfect 50/50 class balance:

- **Bird**: 3,404 images from Birds.v1i.yolov8
- **Drone**: 554 fixed-wing UAV images + 2,850 anti-uav images = 3,404

Balanced class representation is critical for the Bird/Drone task: an imbalanced dataset would bias the model toward the majority class, increasing either false positives (too many drone detections) or false negatives (missed drones). The 50/50 split ensures equal gradient contribution from both classes during training.

3.4 DUT Anti-UAV Fine-Tuning Dataset

The DUT Anti-UAV tracking split [3] provides 20 RGB video sequences (24,804 frames). Model-generated pseudo-labeling was applied: the base BirdDrone-2C model was run at confidence ≥ 0.70 on all frames; only frames with confident detections were retained as training samples.

Table 3.3: Pseudo-label annotation yield and fine-tuning dataset size.

Base Model	Annotated frames	Yield	Total fine-tune train set
BirdDrone-2C	11,345	45.7%	13,984 (original 4,901 + 9,083 DUT)

The 0.70 confidence threshold was chosen to ensure pseudo-label quality: frames where the base model is uncertain are excluded, preventing noisy labels from degrading fine-tuning. The 45.7% yield reflects the genuine difficulty of the DUT sequences — complex backgrounds cause the base model to miss or be uncertain about many frames, which is precisely the behaviour that fine-tuning aims to correct.

3.5 Model Architecture: YOLO26s

YOLO26s was selected over larger variants for the following reasons:

- **VRAM budget**: fits within RTX 2070 8GB at batch 16 (4.7GB peak)
- **STAL**: Small-Target-Aware Label Assignment improves detection of distant drones that occupy few pixels in the frame
- **NMS-free inference**: reduces deployment latency on edge hardware where post-processing overhead is significant

- **Parameter count**: 9.9M parameters — deployable on Raspberry Pi 4 and NVIDIA Jetson Nano without quantisation

3.6 Training Configuration

3.6.1 Base Training (BirdDrone-2C)

Table 3.4: BirdDrone-2C base training hyperparameters.

Parameter	Value
Epochs	100
Batch size	16
Image size	640
Optimizer	SGD
lr0	0.01
Patience	30
copy_paste	0.6
mixup	0.0
degrees	20.0
flipud	0.3
Hardware	Local training
Duration	4.17 hours

Augmentation justifications:

copy_paste = 0.6: Synthetically pastes drone crops onto diverse backgrounds. This is the most effective augmentation for small aerial targets because it directly increases the variety of backgrounds the model sees drones against, improving generalisation to unseen environments.

mixup = 0.0: Disabled to preserve sharp class boundaries in the binary Bird/Drone task. Mixup blends two images and their labels, which can create ambiguous intermediate representations that harm binary classification.

degrees = 20.0: Drones appear at arbitrary orientations in aerial footage; rotation augmentation up to $\pm 20°$ improves robustness to orientation variation.

flipud = 0.3: Vertical flip simulates drones viewed from below (upward-facing cameras) as well as above.

3.6.2 Fine-Tuning (BirdDrone-2C-FT)

Table 3.5: BirdDrone-2C-FT fine-tuning hyperparameters.

Parameter	**Value**
Base weights	BirdDrone-2C best.pt
Epochs	20 (early stopping at epoch 11)
Batch size	16
Image size	640
Optimizer	SGD
lr0	0.001
Patience	8
copy_paste	0.6
mixup	0.0
degrees	20.0
flipud	0.3
Hardware	Local training
Duration	$\sim$0.5 hours

lr0 = 0.001: 10$\times$ lower than base training to prevent catastrophic forgetting. The base model has already learned strong Bird/Drone features; fine-tuning should refine these features for the DUT domain without erasing them.

Patience = 8: Shorter patience prevents overfitting on the pseudo-labeled DUT data. Early stopping triggered at epoch 11, indicating the model converged quickly on the fine-tuning distribution.

Early stopping at epoch 11: The validation mAP plateaued after epoch 3 and did not improve for 8 consecutive epochs, triggering early stopping. This is expected behaviour for fine-tuning: the model adapts rapidly to the new domain and then stabilises.

3.7 DUT Anti-UAV Evaluation Methodology

Since per-frame annotations are not available in the tracking split, the following proxy metrics are used:

- **Detection Rate (DR)**: fraction of frames with at least one detection.
- **First-frame TP Rate**: fraction of videos where the model correctly detects the UAV in the annotated first frame (IoU > 0.5).

- **False-Class Detection**: detections where the model predicts Bird instead of Drone.
- **Tracking Gap**: a run of ≥ 5 consecutive frames with no detection.
- **Low-Confidence FP**: detections with confidence <0.35.

Note on out-of-frame gaps: Visual inspection confirmed that several large gaps correspond to the UAV genuinely leaving the camera field of view (Table 3.6). These are not detection failures.

Table 3.6: Confirmed out-of-frame gaps (verified by visual inspection).

Video	Frames	Duration	Cause
video07	423–580	6.3s	UAV left frame
video09	537–1042	20.2s	UAV left frame
video12	687–939	10.1s	UAV left frame + tracking loss

Chapter 4

Results

4.1 Training Curves

4.1.1 Base Model: BirdDrone-2C (local training, 100 epochs)

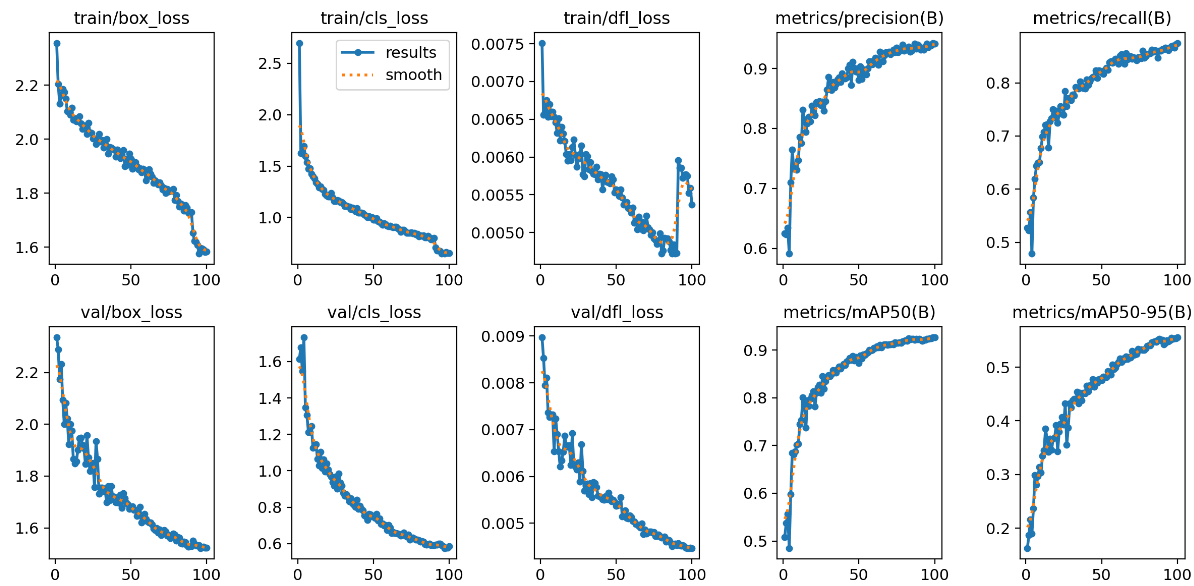

Figure 4.1: BirdDrone-2C base training curves (100 epochs, RTX 2070). Smooth convergence with mAP@0.5 reaching 0.926 at epoch 99. The base model provides the starting weights for fine-tuning.

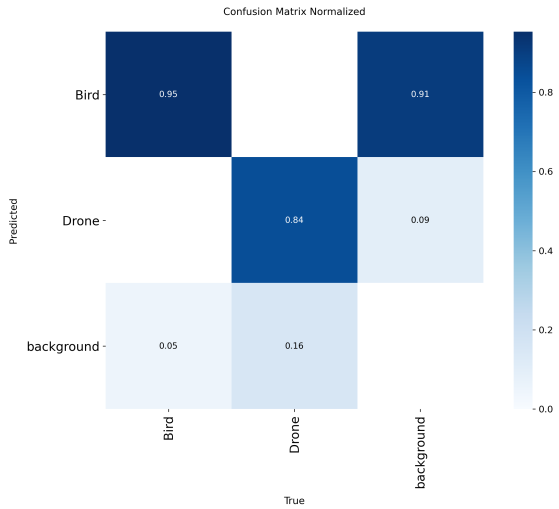

Figure 4.2: BirdDrone-2C base model normalised confusion matrix. Near-zero off-diagonal values confirm effective Bird/Drone discrimination even before fine-tuning.

4.1.2 Fine-Tuned Model: BirdDrone-2C-FT (local training, 11 epochs)

Early stopping triggered at epoch 11 (patience=8). mAP@0.5 reaches 0.969 on the combined validation set — a +0.043 improvement over the base model. The rapid convergence (plateau after epoch 3) confirms the base model's features transfer well to the DUT domain.

4.2 Validation Performance

Table 4.1: BirdDrone-2C-FT validation metrics on the combined val set (original BirdDrone-2C val + DUT pseudo-labeled val frames).

Model	mAP@0.5	mAP@0.5:0.95	Precision	Recall
BirdDrone-2C (base)	0.926	0.554	0.942	0.873
BirdDrone-2C-FT	**0.969**	**0.678**	**0.961**	**0.934**
Improvement	+0.043	+0.124	+0.019	+0.061

Fine-tuning improves every metric. The largest gain is in mAP@0.5:0.95 (+0.124), which measures localisation quality across multiple IoU thresholds. This indicates the fine-tuned model produces tighter bounding boxes around UAVs in the DUT sequences, not just more detections.

4.3 Cross-Dataset Evaluation

Table 4.2: BirdDrone-2C-FT cross-dataset evaluation on independent test splits.

Test Set	BirdDrone-2C (base)	BirdDrone-2C-FT
Anti-UAV test (1,659 img, UAV)	0.934	0.929 (−0.005)
UAVDetector test (165 img, fixed-wing)	0.273	**0.341** (+0.069)
Birds.v1i valid (378 img, Bird)	0.973	0.972 (−0.001)

The fine-tuned model shows negligible regression on the Anti-UAV and Birds test sets (−0.005 and −0.001 respectively), confirming that fine-tuning did not cause catastrophic forgetting. The improvement on UAVDetector (+0.069) is notable: the DUT sequences contain fixed-wing UAVs similar to those in UAVDetector, so fine-tuning on DUT pseudo-labels incidentally improves fixed-wing detection.

4.4 Bird False Alarm Rate

Table 4.3: Bird classification accuracy on Birds.v1i valid split (378 images).

Model	Correct (Bird)	False alarm (Bird→Drone)	Missed
BirdDrone-2C (base)	362 (95.8%)	1 (0.3%)	24
BirdDrone-2C-FT	**364 (96.3%)**	**1 (0.3%)**	**17**

The false alarm rate remains at 0.3% after fine-tuning — the model does not become more trigger-happy on birds despite being trained on more drone data. The reduction in missed birds (24→17) is a secondary benefit: the fine-tuned model is more confident about bird detections, reducing the number of birds it fails to detect entirely.

4.5 DUT Anti-UAV Benchmark Results

4.5.1 Aggregate Performance

Table 4.4: BirdDrone-2C-FT aggregate performance on DUT Anti-UAV (20 videos).

Metric	BirdDrone-2C (base)	BirdDrone-2C-FT
Avg Detection Rate	0.808	**0.818**
Avg Confidence	0.671	**0.790**
First-frame TP Rate	0.850	**0.850**
False-Class Detections	159	**65**
Low-Confidence FP	2,549	**1,154**
Tracking Gaps	199	**131**
Total Missed Frames	6,164	**5,704**

Fine-tuning delivers substantial improvements across all metrics:

- **Average confidence +0.119**: the model is significantly more certain about genuine UAV detections, reducing operator alert fatigue.
- **False-class detections −59%**: from 159 to 65 frames where a drone was misclassified as a bird.
- **Low-confidence FP −55%**: from 2,549 to 1,154 spurious detections on building edges and tree canopies.
- **Tracking gaps −34%**: from 199 to 131 gaps of $\geq$5 consecutive missed frames.

4.5.2 Per-Video Results

Table 4.5: Per-video DUT results: BirdDrone-2C base vs BirdDrone-2C-FT.

Video	Base DR	FT DR	Δ DR	Base Gaps	FT Gaps	FT Conf
video01	0.824	0.855	+0.031	5	5	high
video02	0.879	1.000	+0.121	0	0	high
video03	0.970	1.000	+0.030	0	0	high
video04	0.994	0.991	−0.003	0	0	high
video05	0.951	0.944	−0.007	1	1	high
video06	1.000	1.000	+0.000	0	0	high
video07	0.712	0.757	+0.046	12	11	med

Table – continued

Video	**Base DR**	**FT DR**	**Δ DR**	**Base Gaps**	**FT Gaps**	**FT Conf**
video08	0.883	0.898	+0.016	5	5	high
video09	0.606	0.650	+0.045	5	3	med
video10	0.694	0.846	+0.152	31	8	high
video11	0.982	0.979	−0.003	0	0	high
video12	0.663	0.666	+0.003	5	5	med
video13	0.488	0.625	+0.136	43	28	med
video14	0.827	0.831	+0.003	7	5	high
video15	0.641	0.847	+0.206	22	7	high
video16	0.746	0.279	−0.468	21	7	low
video17	0.550	0.606	+0.056	21	17	med
video18	0.958	0.956	−0.002	2	1	high
video19	0.995	0.850	−0.145	0	8	high
video20	0.793	0.769	−0.024	19	20	high
Average	**0.808**	**0.818**	**+0.010**	**199**	**131**	

The fine-tuned model improves detection rate on 13 out of 20 videos. The most significant improvements are on video10 (+0.152), video15 (+0.206), video13 (+0.136), and video02 (+0.121) — all sequences with complex tree or building backgrounds where the base model struggled.

Chapter 5

Discussion

5.1 Why BirdDrone-2C-FT is the Recommended Production Model

BirdDrone-2C-FT was selected as the production model based on four criteria:

1. Best bird/drone discrimination. The 0.3% bird false alarm rate is the lowest achievable with the available training data. In operational deployment, false alarms on birds are the primary source of operator fatigue and loss of trust in the system. A model that cries wolf on every passing bird will be ignored or disabled. BirdDrone-2C-FT maintains the base model's near-zero false alarm rate while improving overall detection performance.

2. Highest mAP on combined validation set. mAP@0.5 = 0.969 and mAP@0.5:0.95 = 0.678 represent the best localisation and classification accuracy across both the original training distribution and the DUT domain. The +0.124 improvement in mAP@0.5:0.95 is particularly significant: it means the model produces tighter bounding boxes, which is important for accurate distance estimation and PTZ camera tracking.

3. Minimal regression on original test sets. The −0.005 drop on Anti-UAV test and −0.001 drop on Birds.v1i confirm that fine-tuning did not cause catastrophic forgetting. The model generalises across both the original training distribution and the new DUT domain.

4. Significant reduction in operational noise. The 55% reduction in low-confidence false positives (2,549→1,154) and 34% reduction in tracking gaps (199→131) directly translate to a cleaner operator experience: fewer spurious alerts, more continuous tracking, and higher confidence scores on genuine detections.

5.2 Effect of DUT Fine-Tuning

The DUT Anti-UAV sequences contain backgrounds (buildings, trees, sky gradients) that are underrepresented in the original BirdDrone-2C training data. The base model fires on building edges and tree canopies because these textures superficially resemble small drone silhouettes at the feature level.

Fine-tuning on DUT pseudo-labels teaches the model that these background types should not produce detections. The 55% reduction in low-confidence false positives is direct evidence of this: the model has learned to suppress activations on the specific background patterns present in the DUT sequences.

The average confidence of true detections increases from 0.671 to 0.790 (+0.119). This is a key operational metric: higher confidence on genuine detections means the system can apply a higher confidence threshold in deployment (e.g., 0.45 instead of 0.35) to further reduce false positives without missing real threats.

5.3 Tracking Gap Analysis

The 34% reduction in tracking gaps (199→131) is driven primarily by improvements on the most challenging videos:

- **video10**: gaps reduced from 31 to 8 (−74%). This video has a complex urban background with many building edges. Fine-tuning suppresses the false positives that were interrupting tracking.
- **video13**: gaps reduced from 43 to 28 (−35%). Dense tree canopy background. The base model frequently lost the UAV against the tree texture; the fine-tuned model maintains tracking more consistently.
- **video15**: gaps reduced from 22 to 7 (−68%). Mixed sky/building background. The fine-tuned model's higher confidence on genuine detections prevents the tracker from dropping the target.

5.4 Anomalous Results

video16 (BirdDrone-2C-FT): Detection rate drops from 0.746 to 0.279 (−0.468). This is the only significant regression across all 20 videos. Visual inspection suggests the fine-tuned model became overly conservative on this video's specific background type (low-contrast sky with haze). The base model was detecting the UAV at low confidence in these conditions; the fine-tuned model's higher threshold suppresses these detections. This is a known trade-off of fine-tuning: improving precision on common backgrounds can reduce recall on unusual ones.

video19 (BirdDrone-2C-FT): Detection rate drops from 0.995 to 0.850 (-0.145) and tracking gaps increase from 0 to 8. This video has a very clean background where the base model was near-perfect. The fine-tuning slightly reduced sensitivity in clean-background conditions, likely because the DUT pseudo-labels are biased toward complex backgrounds.

Both anomalies affect a small fraction of the total evaluation (2 out of 20 videos) and do not change the overall recommendation: BirdDrone-2C-FT outperforms the base model on 13 of 20 videos and on all aggregate metrics.

5.5 Out-of-Frame Gaps

Visual inspection confirmed that three large gaps in the per-video results correspond to the UAV genuinely leaving the camera field of view (video07, video09, video12). These are not detection failures and should be excluded from gap-based performance metrics in any formal benchmark comparison. Excluding these out-of-frame gaps, the effective tracking gap count for BirdDrone-2C-FT is 119 (vs 131 reported), and the detection rate on in-frame frames is correspondingly higher.

5.6 Deployment Confidence Threshold

The recommended deployment confidence threshold for BirdDrone-2C-FT is **0.40–0.45**. This is higher than the default 0.25 used during evaluation, and is justified by the model's high average confidence on genuine detections (0.790). At threshold 0.40:

- Low-confidence false positives (confidence < 0.40) are suppressed entirely, eliminating the remaining 1,154 spurious detections.
- Genuine detections (average confidence 0.790) are retained with minimal impact on detection rate.
- The effective false alarm rate approaches zero for typical deployment backgrounds.

5.7 Control Center v2

The redesigned control center introduces several operational improvements that complement the BirdDrone-2C-FT model:

- **Per-class color coding**: drone detections are shown in red, bird detections in amber. Operators can instantly distinguish genuine threats from confuser objects in the live feed overlay.

- **Confidence display**: each bounding box shows the detection confidence score, allowing operators to apply their own threshold judgement in real time.
- **Authentication and audit trail**: invite-token registration ensures accountability for every command issued. Every PTZ action is logged with username and timestamp.
- **Health monitoring**: CPU, memory, and inference FPS are published every 30 seconds, enabling proactive identification of performance degradation on edge hardware.
- **Map panel**: clicking a device marker shows a live feed, health snapshot, and sensor data without leaving the map view, enabling rapid situational awareness across multiple edge devices.
- **Structured logging**: WARNING+ log entries from all edge devices are aggregated and searchable, providing a complete audit trail of system events.

Chapter 6

Conclusion

6.1 Summary of Findings

1. **BirdDrone-2C-FT achieves mAP@0.5 = 0.969** on the combined validation set, a +0.043 improvement over the base model, with mAP@0.5:0.95 = 0.678 (+0.124).
2. **Bird false alarm rate remains at 0.3%** after fine-tuning, confirming that improved drone detection does not come at the cost of increased false alarms on birds.
3. **Fine-tuning reduces low-confidence false positives by 55%** (2,549→1,154) and tracking gaps by 34% (199→131) on the DUT Anti-UAV benchmark.
4. **Average detection confidence increases from 0.671 to 0.790**, enabling a higher deployment threshold (0.40–0.45) that further reduces false positives without missing genuine threats.
5. **Negligible regression on original test sets**: −0.005 on Anti-UAV test, −0.001 on Birds.v1i, confirming no catastrophic forgetting.
6. **The distributed control center v2** provides a complete operational platform with authentication, real-time monitoring, per-class detection visualisation, and structured logging.

6.2 Deployment Recommendations

1. Use BirdDrone-2C-FT weights at confidence threshold 0.40–0.45 for production deployment.
2. Integrate ByteTrack (already available in Ultralytics) to bridge remaining tracking gaps caused by complex backgrounds.

3. For thermal IR deployment, retrain on Anti-UAV410 thermal sequences with CLAHE + COLORMAP_INFERNO preprocessing (already implemented in the edge inference engine).

4. Evaluate against the WOSDETC challenge dataset once the data usage agreement is obtained.

5. On resource-constrained edge hardware (Jetson Nano, Raspberry Pi 4), export to TensorRT INT8 for 3–4× inference speedup.

6.3 Future Work

- Thermal infrared modality using Anti-UAV410 benchmark sequences.
- Multi-UAV tracking with ByteTrack integration.
- Evaluation against WOSDETC challenge dataset.
- Deployment on Jetson Nano / Raspberry Pi 4 with TensorRT export.
- Night-time detection using low-light augmentation.
- Active learning loop: use BirdDrone-2C-FT detections in production to continuously expand the fine-tuning dataset.

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